Satellite-Aided Low-Altitude UAV Service Migration With Semantic Extraction and Generated Graphs

Qiongqiong Jia, Jie Zheng, Ling Gao, Jinping Niu, *Member, IEEE*, Rui Cao, and Jie Ren, *Member, IEEE*

***Abstract*— Satellite-aided low-altitude UAV networks utilize satellites to support low-flying drones with communication and navigation for tasks like sensing, surveillance, and delivery. A critical challenge is offloading UAV communication traffic and migrate services to satellites during network congestion. The decentralized structure, changing link conditions, and limited local visibility make it hard to coordinate service migration and request routing in such dynamic environments. To address these challenges, we propose a semantic graph-based multi-agent reinforcement learning (MARL) framework for satellite-aided UAV networks. We formulate service migration and routing as a semantic graph optimization problem, with the objectives of reducing communication delay and increasing network throughput. The framework incorporates two key components: a cyclic message-passing model that enables nodes to infer global network states from limited local observations, and a discrete denoising diffusion model that generates realistic, and dynamic topologies. Our framework leverages semantic feature extraction to further enhance decision-making in routing and service placement. Extensive simulations show that our approach achieves significant reductions in average transmission delay and improvements in the network throughput.**

***Index Terms*— Low-altitude economy, satellite-UAV network, service migration, multi-agent reinforcement learning, semantic extraction graph representation.**

I. Introduction

THE burgeoning low-altitude economy (LAE), powered by sophisticated uncrewed aerial vehicle (UAV) operations, is profoundly reshaping the demands on modern wireless network infrastructure [1]. The highly concentrated LEO satellite-terrestrial integrated network is regarded as a prospective architecture for 6G systems [2]. This architecture can ensure the continuous end-to-end delivery of ultra-distant real-time services. These services involve long-distance data transmission and are sensitive to delays [3]. Specifically, UAVs bring unique advantages to the wireless network ecosystem. In LEO satellite-UAV edge computing networks, UAVs can bridge the communication gap in regions with complex topography. This contributes to worldwide wireless coverage and providing services for mobile subscriber terminals [4]. Thus, the combination of UAVs and satellites is expected to meet the diverse communication needs of global users.

Service migration and request routing are crucial for optimizing network service performance [5], [6] and dealing with satellite and UAV overload [7]. However, the integrated satellite-UAV network is large-scale, heterogeneous, and highly dynamic, making efficient resource orchestration a formidable task [8]. In addition, the different requirements of low latency and high bandwidth services make it challenging to make optimal decisions in service migration and request routing. Although some studies, such as that presented in [9], have proposed solutions for joint delay and completion rate optimization in integrated space-air-ground networks using centralized control mechanisms. A cooperative computation offloading strategy was proposed for UAV-aided MEC systems. This strategy employs user trajectory prediction for initial deployment to minimize energy consumption and path loss [10]. Graph neural networks (GNNs) are designed for graph-structured data. They enable each node aggregate features of itself and the whole graph via interactions with neighbors. Distributed agents with GNN can be used to achieve efficient routing decisions in mega LEO satellite constellations, it need to explore to handle the dynamic changes in network topology after the addition of UAVs [11]. These methods may suffer from high communication overhead and limited scalability. Semantic mobile networks have attracted considerable attention in recent research [12]. A semantic-aware coding and routing joint mechanism was proposed by leveraging semantic correlations and link status to significantly improve data delivery performance in challenging satellite networks [13].

To address these challenges, this article proposes an innovative method that integrates semantic extraction and recurrent message passing into multi-agent reinforcement learning (MARL). MARL is a paradigm where multiple agents interact in a shared environment, each learning an optimal strategy. Through trial-and-error and iteration, they gradually gain the ability to select optimal actions under different states. Furthermore, diffusion models excel in graph generation by

Received 26 May 2025; revised 5 September 2025 and 7 November 2025; accepted 8 December 2025. Date of publication 18 December 2025; date of current version 20 January 2026. This work was supported in part by the National Key Research and Development Program of China (2023YFC3304605), National Natural Science Foundation of China (62406247 and 62372281), Shaanxi Province 2024 Innovation Capacity Support Program (2024ZC-KJXX-014). The associate editor coordinating the review of this article and approving it for publication was L. Zhu. *(Corresponding author: Jie Zheng.)*

Qiongqiong Jia, Jie Zheng, Ling Gao, Jinping Niu, and Rui Cao are with the State-Province Joint Engineering and Research Center of Advanced Networking and Intelligent Information Services, College of Computer Science, Northwest University, Xi'an, Shaanxi 710127, China (e-mail: 202333091@stumail.nwu.edu.cn; jzheng@nwu.edu.cn; gl@nwu.edu.cn; jinpingniu@nwu.edu.cn; cr@nwu.edu.cn).

Jie Ren is with the School of Computer Science, Shaanxi Normal University, Xi'an 710119, China (e-mail: renjie@snnu.edu.cn).

Digital Object Identifier 10.1109/TCCN.2025.3645425

2332-7731 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.

iteratively refining probabilistic representations of network topologies [14], [15], [16], delivering many unique advantages. We use a discrete denoising diffusion model to generate diverse network topologies that can better reflect the dynamic and complex characteristics of UAV-satellite networks [17], [18]. By extracting semantic features from network states, our method can more comprehensively capture the underlying characteristics of the network. The recurrent message-passing mechanism enables agents to gradually obtain approximate global graph observations, overcoming the limitation of restricted neighborhood observations in traditional methods. Our contributions are as follows:

- By integrating semantic extraction, we can capture the semantic information of network states, such as structural features, satellite node attribute features, and packet-related features. This provides a more intelligent foundation for service migration and routing decisions, as it enables a better understanding of the dynamic network environment with lower overhead.
- The recurrent message-passing mechanism enables agents to gradually obtain approximate global graph observations. This addresses the limitation of restricted neighborhood observations in traditional methods, enabling agents to make more informed decisions in dynamic network environments.
- The generation of diverse network topologies using the discrete denoising diffusion model provides a more comprehensive and realistic basis for evaluating and optimizing service migration and routing strategies. It helps to better simulate the actual scenarios of satellite-UAV networks, where the network load is dynamically changing.
- Through extensive simulation experiments, we demonstrate that the proposed method can effectively reduce the average transmission delay and increase the throughput, thereby enhancing the efficiency of service migration and request routing in satellite-UAV networks.

II. Related Work

In the section, we present an overview of related works, including service migration and routing in satellite and UAV networks, as well as the applications of MARL in satellite and UAV networks.

A. Service Migration and Routing in UAV-Satellite Networks

The UAV-satellite network, as a cutting-edge integrated 6G communication framework, is increasingly prominent in the LAE. Satellites, leveraging their vast coverage, support UAV activities by enabling seamless collaboration, which enhances user communication and computing quality through agile deployment and localized services. Chen et al. [19] formulated a joint problem of service migration and power control in energy-harvesting LEO satellite networks to minimize end-to-end task offloading latency. Li et al. [20] proposed the online service deployment to dynamically select service nodes and migrate service entities in mega-LEO satellite constellations. Liu et al. [21] formulated an issue focused on optimizing unbalanced task offloading and vehicle mobility. For the UAV-assisted Internet of Vehicles, they applied the multi-agent deep deterministic policy gradient algorithm. Lin et al. [22] put forward the dual timescale twin delayed deep deterministic policy gradient algorithm to optimize computational offloading in a mobile edge computing infrastructure aided by numerous UAVs. Zhao et al. [23] designed a cooperative system comprising several UAVs and edge clouds, and formulated a Markov decision process to solve the task offloading. Li et al. [24] used deep federated learning to optimize the deployment of UAVs and resource allocation in order to relieve the pressure on cellular networks. Zhang et al. [25] proposed a scheme for energy-saving cooperative offloading for varied tasks in edge computing for satellite networks, constructed a collaborative service architecture, and reduced energy usage in low-earth orbit satellites. Jointly optimizing the migration of services and routing within MEC systems has been studied to minimize response delay through deployment strategies that use rounding methods alongside adaptive request routing mechanisms [26]. This research has been extended to include multidimensional constraints such as storage and asymmetric bandwidth [27]. It has also been further advanced to tackle long-term reconfiguration cost budgets through dynamic frameworks and approximation algorithms [28]. Thus, we investigate the service migration and routing problem in MEC systems, specifically addressing the unique challenges introduced by satellite-assisted low-altitude UAV scenarios. In this context, dynamic characteristics such as high mobility of satellites and UAVs, time-varying network topologies, and heterogeneous delay-bandwidth constraints significantly complicate the joint optimization of service migration and routing strategies. This complexity is particularly pronounced when aiming for lower overhead and decentralized implementation.

B. MARL for Satellite-UAV Networks

MARL methods were employed to explore solutions from multiple dimensions in the critical field of service migration and offloading in satellite and UAV networks. Houidi et al. [29] utilized deep reinforcement learning (DRL) based on the actor-critic architecture to optimize load balancing, with a focus on network quality of service (QoS) parameters. Song et al. [30] proposed a trusted load-distributing routing strategy based on multiple agents dueling dual deep Q-networks, which incorporated node trust values and latency constraints within 6G integrated space-air-ground networks. The hierarchical deep reinforcement learning load-balancing algorithm was proposed to enhance the capacity of multi-domain software-defined networks [31]. You et al. [32] formulated a packet routing structure based upon MARL, employing long short-term memory RNNs for decision-making, effectively reducing packet delivery time. Zhang et al. [11] characterized the routing problem in LEO satellite systems through the lens of a Markov decision process to achieve distributed intelligent routing decisions, demonstrating significant effectiveness in latency control and network dynamic adaptation. Wang et al. [33] designed a specialized satellite link mode to address the

Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.

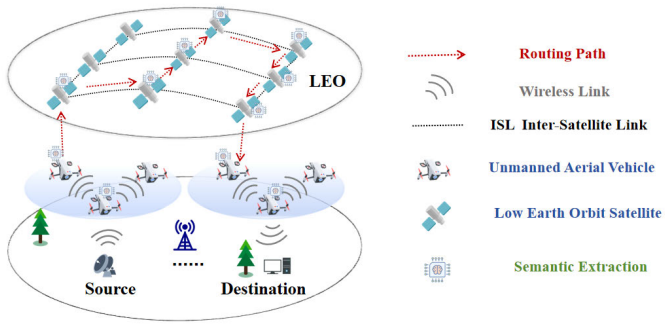


Fig. 1. Service migration with request routing in satellite-aided UAV network.

high energy consumption of laser inter-satellite links (LISLs) and optimized it using MARL, substantially reducing energy consumption and latency. Velasco et al. [34] employed RL to make more intelligent routing decisions in software-defined networks (SDNs). Jiménez et al. [35] reduced the maximum link utilization in SDNs using MARL, where DRL agents at nodes made routing decisions based on local information. Sun et al. [36] addressed UAV base station limitations via collaborative beamforming with virtual antenna arrays, formulating multi-objective optimization and proposing an improved evolutionary multi-objective reinforcement learning algorithm. Existing research on satellite and UAV network has largely focused on resource allocation and link-level optimization under relatively stable network conditions. In contrast, our work targets the semantic-driven graph representation of adaptation and coordination needed in dynamic UAV–satellite environments.

III. Problem Formulation

This paper examines the process of service migration with request routing from UAVs to LEO satellite networks. As illustrated in Figure 1, services in the UAV scenario need to be offloaded to satellites. Due to the rapid mobility of both LEO satellites and UAVs, services can be offloaded and relayed through the UAV layer and multiple LEO satellites via wireless links and inter-satellite links (ISLs). LEO satellites at the same orbital altitude are denoted as $\{1, 2, 3, \ldots, S_{\text{sat}}\}$, while a group of UAVs at the same flight altitude is represented as $\{1, 2, 3, \ldots, U_{\text{uav}}\}$. Each of these is considered a node V_i within the entire satellite network. LEO satellites communicate with each other by establishing ISLs, and UAVs in the high-altitude layer connect through wireless communication. The satellites' movement causes the topology of the LEO satellite network to change dynamically and rapidly, resulting in a time-varying scenario. To capture these rapid changes in network topology, time is divided into multiple slots. We assume that within each time slot, the network topology remains fixed and is modeled as $G(V, E)$, where V and E denote the collections of nodes and communication links in the network, respectively. Each node involved in the routing task within this satellite communication network is equipped with the capability for semantic feature extraction.

We first introduce the key conceptual definitions and then formally state the problems addressed in this work as follows:

1) Link state between satellites: Let n_{ij}^{sat} represent the number of data packets on the link from satellite V_i to satellite V_j within the satellite network, and g_{ij}^{sat} denote the size of these data packets on the corresponding link. For any two satellite nodes V_i and V_j $(i, j \in \{1, \ldots, S_{\text{sat}}\})$ in the satellite network, the link state of the satellite network, denoted as L_{sat}, can be expressed as:

$$L_{sat} = (n_{ij}^{sat}, g_{ij}^{sat}), i, j \in \{1, \cdots, S_{sat}\}. \tag{1}$$

2) Local state observation: For node V_i, consider the local state observation $O_{\text{local}}(V_i)$ of its x-hop neighbors. Let n_{ij}^k be the number of data packets on the link from V_i to its k-hop neighbor V_j $(k \leq x)$, and g_{ij}^k be the size of the data packets on this link. Additionally, let n_i and g_i represent the number and size of data packets of node V_i itself, respectively. Then:

$$O_{\text{local}}(V_i) = \{(n_{ij}^k, g_{ij}^k)\}_{j \in N_x(V_i)} \cup \{(n_i, g_i)\}, \tag{2}$$

where $N_x(V_i)$ is the set of x-hop neighbors of node V_i.

3) Global state: Let $m(V_i)$ represent the local observation of node V_i. After k iterations and t time steps, the global state H can be represented as:

$$H = \sum_{t=1}^{T} \sum_{k=1}^{K} \sum_{V_i \in V} m(V_i)^{(k,t)}, \tag{3}$$

where $m(V_i)^{(k,t)}$ is the local observation of node V_i at the k-th iteration and the t-th time step.

4) Semantic features: Let E be the set of links in the network graph $G(V, E)$. The semantic feature F_{semantic} can be extracted from the link information. For instance, if we consider the link weight w_{ij}, which may relate to the transmission rate of the link or the number of data packets, we can employ a neural network-based semantic extraction model f such that:

$$F_{\text{semantic}} = f(\{w_{ij}\}_{(V_i, V_j) \in E}). \tag{4}$$

A. Service Migration Model

The latency of service migration in the satellite-UAV network is a key element affecting the overall system's performance. In our proposed model, for service migration with request routing, the latency from source node s to destination node d can be broken down into several components.

Transmission latency on satellite links T_{sat}: In the satellite network, the transmission latency T_{ij}^{sat} of the satellite link from node V_i to V_j is determined by the data packet size g_{ij}^{sat} and the transmission rate r_{ij}^{sat} of the link. It can be expressed as:

$$T_{ij}^{sat} = \frac{g_{ij}^{sat}}{r_{ij}^{sat}}. \tag{5}$$

For a service path $R = \{V_s, V_{i_1}, \cdots, V_{i_n}, V_d\}$ passing through satellite nodes, the total transmission latency on satellite links is $T_{sat} = \sum_{k=0}^{n-1} T_{i_k, i_{k+1}}^{sat}$, where $V_{i_0} = V_s$ and $V_{i_n} = V_d$. The nodes V_{i_k} come from the set of satellite nodes $\{1, \cdots, S_{sat}\}$ defined in the system model.

Transmission latency on UAV links T_{uav}: In the UAV network, for the link between a UAV and a satellite, if the

Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.

data packet size is g_{ij}^{uav} and the transmission rate is r_{ij}^{uav}, then the transmission latency T_{ij}^{uav} is given by:

$$T_{ij}^{uav} = \frac{g_{ij}^{uav}}{r_{ij}^{uav}}. \tag{6}$$

For a service migration path passing through the UAV layer, the total transmission latency on UAV links is calculated by summing up the latencies of each individual link in the UAV related of the service migration path.

Queuing latency T_{queue}: The queuing latency at node V_i is related to the quantity of data packets n already queued at that node. For a basic $M/M/1$ queuing model, we assumed Poisson arrival and exponential service time, the queuing latency T_i^{queue} is given by:

$$T_i^{queue} = \frac{\lambda_i^2}{\mu_i(\mu_i - \lambda_i)}. \tag{7}$$

where λ_i is the arrival rate of data packets at node V_i and μ_i is the service rate. The arrival rate λ_i can be related to the number of incoming data packets from adjacent nodes, such as local state observation, and the service rate μ_i is related to the transmission rate of the outgoing link of node V_i.

The total queuing latency for the path of service migration with request routing R is $T_{queue} = \sum_{V_i \in R} T_i^{queue}$.

Propagation latency $T_{propagation}$: In this satellite-UAV network, the propagation latency refers to the latency generated when the signal propagates in the transmission medium, such as the space between satellites, the wireless links between UAVs and satellites or ground stations. For the link from node V_i to V_j, its propagation latency $T_{propagation}(V_i, V_j)$ is mainly determined by the physical distance $d(V_i, V_j)$ of the link and the signal propagation speed c, and can be expressed as:

$$T_{propagation}(V_i, V_j) = \frac{d(V_i, V_j)}{c}. \tag{8}$$

For a multi-hop link, the total propagation latency $T_{propagation-total} = \sum_{(V_i,V_j)\in R} \frac{d(V_i,V_j)}{c}$, where R is the service migration with request routing path.

In semantic feature extraction, the latency is significantly affected by model complexity, computing resources, and the scale of input data. For the fully connected neural network f with N hidden layers (each having M neurons), its processing time for input data is proportional to $N \times M \times |E|$. Considering computing resources, with t_0 as the time for one neuron operation per unit computing resource and C as the device's available computing resources, the semantic feature extraction latency T_{semantic} is:

$$T_{\text{semantic}} = \frac{N \times M \times |E|}{C} \times t_0. \tag{9}$$

The total latency T_{total} from the source to the destination is $T_{total} = T_{sat} + T_{uav} + T_{queue} + T_{propagation} + T_{semantic}$.

B. Problem Description

Service migration with request routing is crucial for optimizing network performance and dealing with issues such as UAV overload. We are committed to exploring an optimal service migration with request routing scheme. The goal is to minimize the total cost incurred from the originating node to the target node for service migration and to the destination node for request services, while satisfying a series of constraints.

The total delay here includes satellite link transmission delay, UAV link transmission delay, queuing delay, propagation delay, and semantic feature extraction delay. To accurately model the service migration and request routing problem, we need to consider a series of constraints, which reflect the actual operation situation and resource limitations of the network.

a) Request routing link feasibility: This constraint ensures that the nodes and links on the routing path actually exist and are traversable in the network topology.

$$C1: \begin{cases} v_i \in V, & \forall i = 1, \cdots, m, \\ (v_i, v_{i+1}) \in E, & \forall i = 1, \cdots, m-1, \end{cases} \tag{10}$$

where v_i represents the i-th node in the offloading P_{sd}, the collection of nodes in the network topology diagram is denoted as V, and E stands for the collection of links.

b) Satellite link capacity: For satellite links, the quantity of data packets $n_{v_i v_{i+1}}^{sat}$ and the size of data packets $g_{v_i v_{i+1}}^{sat}$ carried on the link from node v_i to v_{i+1} must meet the carrying capacity limitations of the link itself. That is, $n_{v_i v_{i+1}}^{sat}$ should not exceed the maximum number of packets N_{max}^{sat} that the satellite link can carry, and $g_{v_i v_{i+1}}^{sat}$ should not exceed the maximum packet size G_{max}^{sat} that the satellite link can carry. This constraint holds when v_i and v_{i+1} are satellite nodes.

$$C2: \begin{cases} n_{v_i v_{i+1}}^{sat} \le N_{max}^{sat}, \\ g_{v_i v_{i+1}}^{sat} \le G_{max}^{sat}. \end{cases} \tag{11}$$

c) UAV link capacity: For UAV links, the quantity of data packets $n_{v_i v_{i+1}}^{uav}$ and the size of data packets $g_{v_i v_{i+1}}^{uav}$ carried on the link from node v_i to v_{i+1} also need to meet the carrying capacity limitations of the link. That is, $n_{v_i v_{i+1}}^{uav}$ should not exceed the maximum number of packets N_{max}^{uav} that the UAV link can carry, and $g_{v_i v_{i+1}}^{uav}$ should not exceed the maximum packet size G_{max}^{uav} that the UAV link can carry. This constraint holds when v_i and v_{i+1} involve UAV nodes.

$$C3: \begin{cases} n_{v_i v_{i+1}}^{uav} \le N_{max}^{uav}, \\ g_{v_i v_{i+1}}^{uav} \le G_{max}^{uav}. \end{cases} \tag{12}$$

d) Transmission rate constraint: This constraint ensures that data can meet certain speed requirements during the transmission process.

$$C4: R(v_i, v_{i+1}) \ge R_{min}, \text{ where } R(v_i, v_{i+1}) = \begin{cases} r_{v_i v_{i+1}}^{sat}, \\ r_{v_i v_{i+1}}^{uav}, \end{cases} \tag{13}$$

where $R(v_i, v_{i+1})$ is the actual data transmission rate from node v_i to v_{i+1}, R_{min} is the minimum required transmission rate, $r_{v_i v_{i+1}}^{sat}$ is the data transfer rate of the satellite link, and $r_{v_i v_{i+1}}^{uav}$ is the data transfer rate of the UAV link.

e) Semantic feature extraction resource constraint: This constraint aims to ensure that the semantic feature extraction

Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.

process is carried out within a reasonable resource range.

$$C5: \begin{cases} C_{v_i} \leq C_{max}, \\ |E_{v_i}| \leq |E_{max}|, \end{cases} \tag{14}$$

where C_{v_i} is the computational resource used for semantic feature extraction at node v_i, C_{max} is the upper limit of available computational resources at node v_i, $|E_{v_i}|$ is the size of the link set related to node v_i, and $|E_{max}|$ is the maximum input scale of the semantic extraction model.

f) Feasibility constraint for service migration: This constraint ensures that the target node for service migration has the ability to receive and process the migrated service.

$$C6: \quad Cap_{v_{i+1}} \geq Dem_{service}, \forall i = 1, \cdots, m-1, \tag{15}$$

where $Cap_{v_{i+1}}$ is the service processing capacity of node v_{i+1}, and $Dem_{service}$ is the requirement of the migrated service.

g) Integrity constraint for service offloading: This constraint guarantees that all the service data that needs to be offloaded can be transmitted completely.

$$C7: \quad \sum_{i=1}^{m-1} Data_{offload}(v_i, v_{i+1}) = TotalData_{offload}, \tag{16}$$

where $Data_{offload}(v_i, v_{i+1})$ is the amount of service data offloaded from node v_i to node v_{i+1}, and $TotalData_{offload}$ is the total amount of service data that needs to be offloaded.

h) Resource allocation constraint: This constraint ensures that during the process of service migration with request routing, the resources allocated to services by a node do not exceed its available resources.

$$C8: \quad RA_{used}(v_i) \leq RA_{available}(v_i), \forall i = 1, \cdots, m, \tag{17}$$

where $RA_{used}(v_i)$ is the amount of resources used by node v_i in service migration and offloading, and $RA_{available}(v_i)$ is the amount of available resources of node v_i.

Let the cost of service migration with request routing scheme be S_{sd}, which represents a series of decisions overhead from the source node s to the service offloading destination node d, including determining which services need to be migrated, to which nodes they are migrated, and the data can be transmitted. We set the optimization objective as minimizing the total cost function $Cost(S_{sd})$:

$$\begin{aligned} P: \quad & S_{sd} = \underset{\{S_{sd}\}}{\arg\min} \quad Cost(S_{sd}) \\ Cost(S_{sd}) = & \sum_{k=1}^{m-1} \Big(T_{sat}(v_k, v_{k+1}) + T_{uav}(v_k, v_{k+1}) \\ & + T_{queue}(v_k) + T_{propagation}(v_k, v_{k+1}) \\ & + T_{semantic}(v_k) + C_{mig}(v_k, v_{k+1}) \\ & + C_{offload}(v_k) \Big) \\ \text{s.t. } & C1, C2, C3, C4, C5, C6, C7, C8. \end{aligned} \tag{18}$$

where C_{mig} is the service migration cost, which can be expressed as follows:

$$C_{\text{mig}}(v_k, v_{k+1}) = \frac{D_{\text{mig}}}{r} + \frac{Dem}{Res(v_{k+1})} + \frac{Dem}{Res(v_k)}, \tag{19}$$

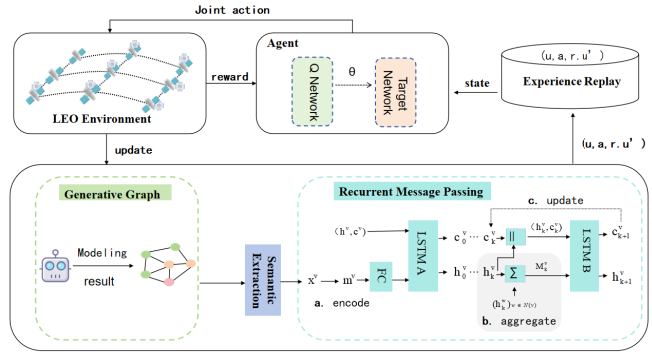


Fig. 2. MARL strategy with semantic extraction and generated graphs.

$$C_{\text{offload}} = \frac{K \cdot |V_k|}{Res(v_k)} + \frac{Data_{\text{offload}} \cdot \rho}{r}, \tag{20}$$

where D_{mig} denotes the migration data volume, Dem stands for the service resource demand, and $Res(\cdot)$ represents the node available resources. $C_{offload}(v_k)$ is the cost of service offloading at node v_k. $|V_k|$ denotes the number of neighbor nodes.

IV. Generative Semantic Recurrent Graph Representation Strategy in MARL for Satellite-Aided UAV Network

In this section, we propose a recursive message-passing mechanism to handle the graph structure and combine MARL to train and execute the service migration with routing. Then, we perform semantic extraction graph representation on the network topology and its observations to further reduce latency and increase throughput. The layout of the proposed model is depicted in Figure 2.

A. Generative Network Topology

In this section, the satellite-UAV network is modeled as a graph $G = (V, E)$, where V represents the collection of satellites within the constellation, and E represents the collection of connections between satellites and UAVs. We utilize discrete denoising diffusion algorithm to generate graphs that possess node and edge attributes [37]. Through a discrete diffusion procedure, the graph is incrementally refined, and a graph transformer network is trained to invert this procedure.

The denoising network's core includes 10 graph transformer layers for extracting graph-structured features. Training is set to 100,000 epochs with a batch size of 64, and a cosine noise scheduling strategy is used to better control noise variation during training, improving generated sample quality. The formula is:

$$\bar{\alpha}^t = \cos\left(0.5\pi \frac{(t/T + s)}{(1+s)}\right)^2, \tag{21}$$

where s denotes a hyperparameter that controls the smoothness of the cosine noise decay curve, preventing abrupt drops or rises in noise intensity during the initial training phase.

The forward diffusion process is a discrete state transition process. Noise is applied independently to each node and edge, and the graph is gradually edited by adding or deleting edges

Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.

and changing the categories of nodes or edges. A graph transformer network is employed to invert the diffusion process. This architecture accepts a noisy graph $G^t = (X, E)$ as its input and outputs tensors X' and E' that signify the predicted distribution of the undistorted graph.

We set the quantity of nodes in the satellite network topology generated by diffusion model, while the number of edges is not fixed. Since the network load in UAV and satellite networks is dynamically changing, by repeating this process, the model can generate a variety of network topologies. These topologies reflect different possible states of the satellite-UAV network under dynamic loads, providing more comprehensive scenarios for service migration with request routing.

B. Message Recurrent Passing Mechanism

GNN and neural message passing exhibit certain advantages in graph-based environment applications. However, these approaches have inherent limitations. They assume a centralized perspective, require explicit coordination among all agents, or rely on labeled data to apply supervised learning, making them unsuitable for decentralized MARL scenarios. Graphs in real-world networks are diverse and dynamic. The solutions learned by traditional agents trained on a single graph are difficult to generalize to other graphs. In graph-based environments, MARL focuses on decentralized systems. During execution, agents can only obtain limited local information. While they are capable of directly reacting to local changes, exclusive dependence on local information may lead to suboptimal decision-making. A common approach is to expand the agents' observation range to include information from their direct neighbors, but this will increase the communication overhead. Thus, we propose a recursive method for obtaining graph observations. The fundamental concept lies in leveraging the message-passing architecture of GNNs to propagate local information throughout the network. A novel recursive feedback loop is integrated, which redirects to the GNN input, and this approach is defined as recurrent message passing.

Under this framework, every node $v \in \mathcal{V}$ obtains local node observations m^v sampled from a distribution $\mathcal{M}^v(\cdot|s)$ conditioned on an unknown system state $s \in \mathcal{S}$. Rather than directly feeding these observations into the GNN, each node $v \in \mathcal{V}$ utilizes a differentiable embedding function to incorporate its local observation into the current node state h^v, as expressed by:

$$h_0^v \doteq encode\left(h^v, m^v\right). \tag{22}$$

In the initial step, the node state h_v is initialized to zero and subsequently acts as a recurrent loop between consecutive environment steps.

During iteration k_0, every node transmits its state h_v^k to all direct neighbors $w \in \mathcal{N}(v)$, where $\mathcal{N}(v) = \{w \mid (v, w) \in \mathcal{E}\}$. Each neighbor generates a new state through a two-step process: first, aggregating the incoming node states using an arbitrary differentiable function aggregate^k, and then updating its own state via another differentiable function update^k. A single message-passing iteration is thus defined by this sequence of operations.

Algorithm 1 Distributed State Update With Semantic Feature Extraction

Input: Node v along with its direct neighbors $N(v)$, current state s, previous state of node v denoted as h^v, and the observation of node v, m^v.

Output: The updated state of node v, h_K^v, and the set of intermediate states, H^v.

Extract the semantic features from the local information of node v and denote them as X_0.
Set the node's local observation m^v to the extracted semantic features X_0.
Generate the initial encoded state h_0^v by encoding the previous state h^v and the current local observation m^v.
for $k = 0$ to $K - 1$ do
  Transmit the current state h_k^v of node v to all its adjacent nodes $w \in N(v)$.
  Receive the current states h_k^w from all neighboring nodes $w \in N(v)$.
  Compute the aggregated state M_k^v by applying the aggregation function aggregate^k to the states h_k^w of all adjacent nodes $w \in N(v)$.
  Update node state v to h_{k+1}^v by using the update function update^k with the current state h_k^v and the aggregated state M_k^v.
end for
Construct the set of intermediate states H^v by concatenating the sequence of states $(h_k^v)_k$ and the sequence of neighboring states $(h_k^w)_{w \in N(v), k}$.
return The final updated state h_K^v and the set of intermediate states H^v.

$$M_k^v \doteq aggregate^k\left(\left(h_k^w\right)_{w \in N(v)}\right), \tag{23}$$

$$h_{k+1}^v \doteq update^k\left(h_k^v, M_k^v\right). \tag{24}$$

The pseudocode for recurrent message passing, executed by each node $v \in \mathcal{V}$, is presented in Algorithm 1. A series of $K \in \mathbb{N}$ message-passing iterations occur between consecutive environment steps. Specifically, equations 23 and 24 are iteratively applied until $k + 1 = K$. The agents can leverage the final aggregated state h_v^K and all intermediate node states H_v computed by node v during these iterations, as elaborated in subsequent sections. In the subsequent environment step, the node state is updated to $h_v \leftarrow h_v^K$, and the algorithm recommences. The number of iterations K can be dynamically tuned to align with the learning task's requirements. While increasing K accelerates information propagation across the network, it also amplifies communication overhead.

At each timestep, the environment undergoes a node state update. We assume that every agent i ($i \in I$) is deterministically mapped to a single node $v^i \in V$ per timestep. In addition to its direct environmental observations, agent i receives a local graph observation ψ^i corresponding to its assigned node. This observation is generated by applying a differentiable readout function Ψ to the accumulated node information H^{v^i}

Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.

during the current timestep.

$$\psi^i \doteq \Psi(H^{v_i}). \tag{25}$$

In the most straightforward scenario, Ψ can be defined as the identity function of the latest node state in each iteration, such that $\Psi(H^{v^t}) \doteq h_K^{v^t}$. This means agents receive the current state of their assigned node directly. Access to intermediate node states is essential to enable skip connections and aggregation mechanisms, as utilized in jumping knowledge networks.

C. MARL Strategy With Semantic Extraction and Generated Graphs

Figure 2 illustrates the MARL architecture for integrated satellite-UAV networks. Semantic information is crucial for the optimization of service migration with routing decisions. Combining the previously described generative network topology and message recursive transmission mechanism, this section will elaborate in detail on the MARL strategy with extraction and generated graph representation. The details of this strategy are shown in Algorithm 2.

In this strategy, each node has the ability to extract semantic features. Before transmitting local information to neighbor nodes, a node will first perform a semantic feature extraction operation. It uses its own semantic feature extraction ability to process its local information, such as the semantic information of the local topological graph. The semantic features we define here are as follows.

Structural features: Degree centrality can intuitively reflect the degree of connection between a node and other nodes. For a node $v \in V$ in the network, its degree centrality can be defined as:

$$C_{\text{degree}}(v) = \frac{\deg(v)}{|V| - 1}, \tag{26}$$

where $\deg(v)$ represents the number of neighbors of node v, and $|V|$ is the total number of nodes in the network.

Node attribute features of satellite: The location information, such as latitude and longitude coordinates, orbit type, and energy state of satellite nodes are also important semantic features. The definitions of the attribute features of satellite nodes are as follows.

The latitude and longitude coordinates of satellite node s are (lat_s, lon_s). The relative distance between it and another node t can be calculated by the spherical distance formula:

$$d(s,t) = R \cdot \arccos\Big[\sin(\text{lat}_s)\sin(\text{lat}_t) + \cos(\text{lat}_s)\cos(\text{lat}_t)\cos(\text{lon}_s - \text{lon}_t)\Big], \tag{27}$$

where R is the Earth's radius.

The energy state of satellite node s can be expressed as a normalized variable:

$$e_s = \frac{E_s}{E_{\max}}, \quad (0 \le e_s \le 1). \tag{28}$$

Packet-related features: The local observation information of a node includes the number and size of packets on outgoing edges, as well as the number and total size of packets on the node. For an outgoing edge $(v, w) \in E$ of node v, let $n_{v,w}$ be the number of packets on this edge, and $g_{v,w}$ be the total size of the packets. Then, the outgoing edge load of node v can be quantified as:

$$L_{\text{out}}(v) = \sum_{w \in N(v)} \left(n_{v,w} + \alpha \cdot g_{v,w}\right), \tag{29}$$

where α is a weight coefficient (balancing the influence of quantity and size), and $N(v)$ is the set of neighbor nodes of v.

The above features can be used as components of the link weight w_{ij}:

$$w_{ij} = \beta_1 \cdot C_{\text{degree}}(i) + \beta_2 \cdot \frac{1}{d(i,j)} + \beta_3 \cdot L_{\text{out}}(i) + \beta_4 \cdot e_i. \tag{30}$$

Satellites with close positions and matching orbits tend to form stable connections, which enhances node centrality. Nodes with high centrality undertake more forwarding tasks, have larger outgoing edge loads, and are closely associated with packet-related features. Satellite attributes affect transmission delay and rate, which in turn influence data packet accumulation, while the packet status also exerts a reverse effect on node connection frequency and centrality.

In node, the extracted semantic features is shown as X_0 and uses it as the local node observation m^v. At this time, the current node state h^v is initialized to 0. Then, each node will enter the message iteration process according to the obtained semantic features and its own data information. In this process, as the number of iterations increases, the cumulative messages received by the node will increase, and the node will gradually obtain nearly global information.

Given that the conventional Deep q-network (DQN) encounters specific limitations in complex environments, the agent calculates the Q value according to the current state s_t^i and the local graph observation ψ^i:

$$Q(s_t^i, \psi^i, a^i) = \mathbb{E}\left[R_{t+1}^i + \gamma \max_{a'^i} Q(s_{t+1}^i, \psi'^i, a'^i)\right], \tag{31}$$

where R_{t+1}^i represents the reward that agent i acquires at time $t+1$, while γ denotes the discount factor, which reflects the degree of importance that the agent attaches to future rewards.

We redefine the service migration with request routing problem within the integrated satellite-UAV network as a sequential decision framework grounded in MARL.

Let $\mathcal{I}$ represent the set of agents and $\mathcal{V}$ denote the set of network nodes. Each agent $i \in \mathcal{I}$ is associated with a node $v_i \in \mathcal{V}$. Our goal is to find an optimal policy π^* that minimizes the total cost of service migration with request routing. This problem can be expressed as follows:

$$\pi^* = \arg\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{T-1} \gamma^t r_t(\pi)\right], \tag{32}$$

where π denotes the policy function, which generates an action a_t conditioned on the current state s_t, such that $a_t = \pi(s_t)$. The discount factor $\gamma \in [0, 1]$ serves to balance the significance of immediate rewards and future rewards. $r_t(\pi)$ represents the reward yielded by implementing the policy π at

Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.

time step t. T is the maximum number of time steps within a decision-making cycle, and it satisfies $0 < T \leq T_{max}$, where T_{max} is the preset maximum number of time steps to avoid infinite-loop decision-making.

Meanwhile, the service migration path P_{sd} can be represented as a sequence of actions:

$$P_{sd} = \langle a_0, a_1, \cdots, a_{T-1} \rangle. \tag{33}$$

Subsequently, we specify the state, action, and reward functions of the MARL architecture.

1) State: The state s_t^i of agent i at time step t usually includes local node information, that is, the self-attributes of node v^i; the local graph observation ψ_t^i obtained through the message recursive transmission mechanism. The model considers three types of information: the state h_K^w $(w \in N(v^i))$ of neighbor nodes, the number and size of packets on the links, and the destination node d_{packet} of the packets.

2) Action: The action a_t^i of agent i at time step t represents the service migration with routing decision of node v^i for the current packet. The action space A^i can be defined as the set of all possible next-hop nodes of node v^i, that is:

$$A^i = \{w | w \in N(v^i)\}, \tag{34}$$

where $N(v_i)$ denotes the collection of immediate neighboring nodes of node v_i. When agent i selects the action $a_t^i = w$, it means forwarding the current packet to the neighbor node w. The action selection is associated with the Q-value (Equation (31)) and the ε-greedy strategy.

3) Reward: The reward function r_t^i is used to motivate agents to make decisions that are beneficial to optimizing routing performance. The reward function we designed takes the following factors into comprehensive consideration: if the packet successfully reaches the destination node d at time step t, a positive reward $R_{success}$ is given to encourage the agent to find an effective service migration path. If the destination node d_{packet} of the data packet is reached at time t, then $R_{\text{success}} = K \cdot \frac{1}{T_{\text{total}}}$, where K is the reward coefficient.

We provide corresponding rewards or penalties based on the transmission delay T_{delay} of the data packet at the current time step. A linear function combining propagation delay and queuing delay is employed: $f(T_{\text{delay}}) = \alpha_1 T_{\text{propagation}}(v^i, w) + \alpha_2 T_{\text{queue}}(v^i)$.

To avoid over-loading of certain links, we expect agents to consider the load conditions of links when making service migration decisions. A load balancing reward R_{balance} is defined to incentivize this behavior. The load status of the link between node v^2 and its adjacent node can be measured by calculating the link load rate $\rho_{v^i w}$, which reflects the relative relationship between the current data volume carried by the link and the maximum carrying capacity of the link. The load balancing reward R_{balance} can be defined based on the link load rate ρ as follows:

$$R_{\text{balance}} = 1 - \max\left(\rho_{v^i,w} - \rho_{\text{th}}, 0\right) \tag{35}$$

where ρ_{th} is the load threshold. When $\rho_{v^i,w} \leq \rho_{\text{th}}$, $R_{\text{balance}} = 1$, indicating perfect load balance; otherwise, the reward decreases as the load exceeds the threshold.

Algorithm 2 MARL With Generative Semantic Graph

Input: Replay memory M, action-value function F with weights ω_F, target weights ω_F^*, node-value function G with weights ω_G.

Output: Updated weights ω_F and ω_G.

Initialize M, F with ω_F, ω_F^*,G with ω_G.
for Episode $k = 0, \cdots$ do
  $x_0 \leftarrow 0$
  Obtain s_0, o_0, n_0 by resetting the environment.
  for Time-step $t = 0$ to $T - 1$ do
    The current satellite obtains the graph observation G^* according to Algorithm 1.
    $(x_{t+1}, z_t) \leftarrow G^*(x_t, n_t, s_t; \omega_G)$.
    for Agent $a \in \mathcal{A}$ do
      With probability β, choose a random action u_t^a.
      Otherwise, $u_t^a = \underset{u}{\text{argmax}} F(o_t^a \parallel z_t^a, u; \omega_F)$.
    end for
    Take action u_t, get reward r_t, new state s_{t+1}, new observation o_{t+1}, new node observation n_{t+1}.
    Store $(x_t, x_{t+1}, n_t, n_{t+1}, s_t, s_{t+1}, o_t, o_{t+1}, u_t, r_t)$ in M.
  end for
  $Loss \leftarrow 0$
  for Batch index b in M ($b = 0$ to $B - 1$) do
    if $b = 0$ then
      $x'_b \leftarrow x_b$
    end if
    $(x'_{b+1}, z'_b) \leftarrow G^*(x'_b, n_b, s_b; \omega_G)$.
    $(x'_{b+2}, z'_{b+1}) \leftarrow G^*(x'_{b+1}, n_{b+1}, s_{b+1}; \omega_G)$.
    $E_b^a = \begin{cases} 0, & \text{if agent } a \text{ is done at step } b+1 \\ 1, & \text{otherwise} \end{cases}$.
    $y_b \leftarrow r_b + E_b^a \gamma \underset{u}{\max} F(\hat{O}_{b+1}^a \parallel z'_{b+1}, u; \omega_F^*)$.
    $Loss \leftarrow Loss + (y_b - F(o_b \parallel z'_b, u_b; \omega_F))^2$.
  end for
  Update ω_F and ω_G by gradient descent on $Loss$.
  Update ω_F^*.
end for

The reward function of agent i at time step t can be expressed as:

$$r_t^i = \begin{cases} R_{success}, & \text{at time } t, \\ -f(T_{delay}) + \beta R_{balance}, & \text{otherwise}, \end{cases} \tag{36}$$

where β is a weight coefficient used to adjust the proportion of the link load balancing reward in the total reward.

Complexity Analysis: With parameters defined as: K-episodes, T-time-steps per episode, $|A|$-agents, A-action space dimension,g-function G^* complexity, B-batch size. Single training time-step complexity is$O(|A|(A+g))$, leading to pre-update training complexity $O(K \cdot T \cdot |A|(A+g))$; update stage complexity is$O(B(2g+1))$; discrete denoising diffusion-based generative graph modeling adds$O(T_{gen} \cdot L \cdot (|V| \cdot d^2 + |E| \cdot d))$($T_{gen}$: graph generations, L: Transformer layers, $|V|/|E|$: nodes/edges); semantic feature extraction adds$O(K \cdot T \cdot |A| \cdot (|V| + |E|(1+h)))$.

Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.

V. Performance Evaluation

A. Experiment Settings

To assess the efficiency of our proposed method, we have established a LEO satellite constellation network comprising 20 satellites, evenly distributed across 4 polar orbits (5 satellites per orbit) at an altitude of approximately 500km with an orbital inclination of 98° [11]. And a small-scale UAV swarm consisting of 8 UAVs is set up [36]. To model heterogeneous connectivity, each satellite has a variable number of outgoing edges (ranging from 2 to 6), and in-orbit links have capacities ranging from 20 to 80 Mbps, while cross-orbit links have capacities from 1 to 10 Mbps. The data packets transmitted in the network have sizes randomly generated within the range of $(10, 100)$Kbs. UAVs communicate with satellites via links at (0.1,0.8) Mbps, and each UAV node can establish bidirectional links with 2 LEO satellites within its coverage area, with link statuses dynamically changing as the UAV moves.

We use a discrete denoising diffusion model referred to as Digress [38] to predictively generate noise-free satellite-UAV network topology graphs from noisy inputs. With the generation results of the model, the degree distribution is 1.4, the clustering coefficient is 1.2, and the orbit count is 1.7. The calculation of the indicator values follows the ratio of the squared Maximum Mean Discrepancy (MMD) between the generated graphs and the test graphs to the squared MMD between the training graphs and the test graphs. These metrics are all significantly smaller than the distribution differences of other baseline models (such as GraphRNN, and GRAN). During the service migration training process of the satellite-UAV network, one graph is generated every 50 time steps, with a total of 2.5×10^6 steps in the training process. The algorithm is tested on 1000 generated topology graphs. Time steps refer to discrete temporal units within a network environment that facilitate state updates, decision execution, or information exchange, representing the iterative progression of the reinforcement learning environment.

To observe the satellite-UAV graph environment, we employ a dedicated observation module. This module operates as a wrapper, ensuring it is decoupled from the MARL process. For this graph observation, our scheme utilizes a GNN that integrates both non-recursive and recurrent network architectures.

The intelligent agent architecture employs the DQN, which incorporates a single linear layer. This layer has input and output dimensions of (256, $A+1$). Its function is to map the encoded observations to a discrete action space encompassing $A+1$ actions. Here, $A=3$ is set, and the 4 discrete actions include waiting and choosing out edges.

We have compared the performance for different strategies:

1) Greedy optimal approach: This strategy uses Dijkstra's algorithm, which utilizes a static graph computation to get the shortest path to the target node. The agent guides the learning based on immediate rewards and this strategy depends on the pre-defined the global information fixed by the satellite network.

2) Service migration with local observation: This strategy utilizes the non-recursive graph network structure to obtain local information, on which the agent makes routing decisions.

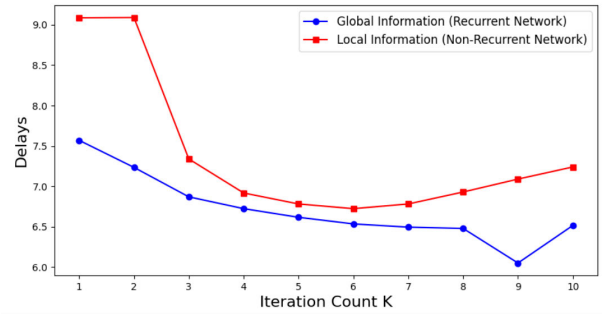


Fig. 3. Average delay comparison at different K.

3) Service migration with global observation: It is utilizing the recursive graph network structure to obtain the global information in the satellite network, but it is not keen on the structural changes of the satellite network.

4) Service migration with keen observation: This scheme relies on global information as well as iteratively capturing the graph state of the previous time step to find the shortest service migration path.

5) Service migration incorporated action masking: To prevent routing loops, an action masking mechanism is added to ensure unnecessary delays on the routing path.

6) Service migration with semantic feature extraction: Semantic extraction of the structure of the satellite network topology is performed to reduce the amount of information data in the global state.

B. Performance Evaluation and Analysis

In order to verify the performance advantages of the cyclic structure, we conducted tests on 10 different values of K under the same traffic background. The experimental results show that our proposed algorithm adopting the recurrent structure can plan the path more efficiently, reducing the transmission delay caused by poor path selection. In the experiment, the unit of throughput is the number of data packets arriving at the destination per time step. The unit of delay is the number of time steps. Reward refers to the average reward per step.

Figure 3 shows that the average latency difference between the recurrent structure and the non-recurrent structure is 1-2 time steps under 10 different K values. Figure 4 shows that the throughput under the recurrent structure outperforms the non-recurrent structure at every iteration. As the number of message iterations increases, both strategies appear to perform better and better upfront in terms of both latency and throughput. However the larger the value of K, the average latency appears to increase and the throughput appears to decrease. This is because as the number of iterations per step increases, more information comes to be exchanged in the network, which in turn increases the consumption of system resources. Therefore, selecting an appropriate number of iterations is crucial to strike a balance between gathering sufficient global information and minimizing communication overhead.

Figure 5 is set to further evaluate the effectiveness of different graph neural network architectures. We designed a shortest path regression problem to verify the adaptability of recurrent and non-recurrent network architectures in the

Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.

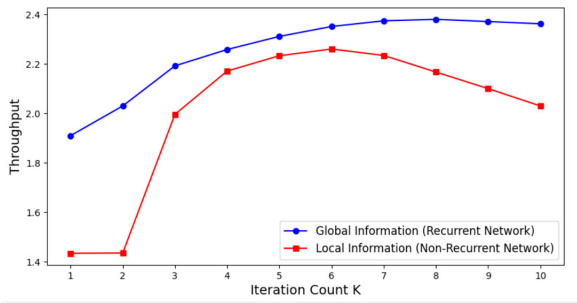


Fig. 4. Throughput comparison at different K.

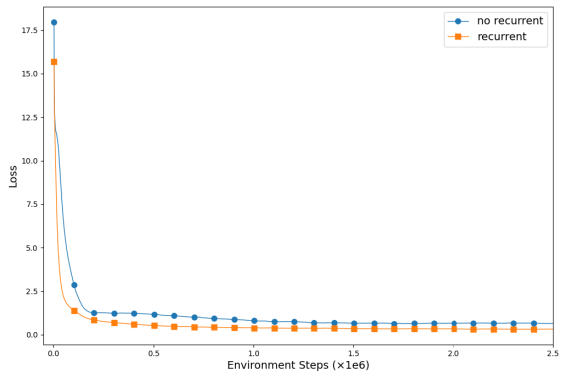


Fig. 5. Loss in recurrent and non-recurrent networks.

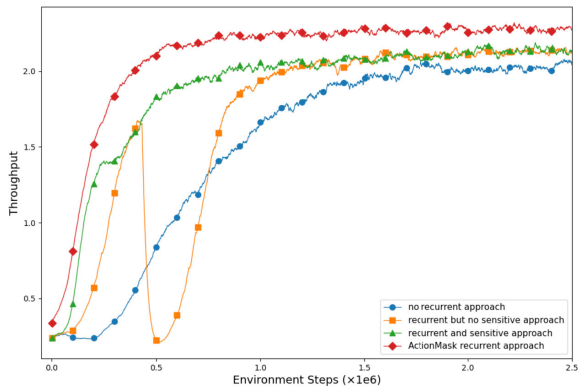


Fig. 6. Comparison of throughput under different routing policies.

service migration environment, where the loss function of this problem is defined as the mean squared error between the predicted distance and the real distance for each source node and destination node. Experimental results show that the recurrent network structure can converge quickly in fewer time steps, thereby verifying the effectiveness of using the recurrent structure to obtain global information.

Figure 6 illustrates the throughput of different service migration strategies. The experimental results show a significant increase in throughput when using a recurrent network to handle graph environment observation compared to a non-recurrent network. In the scenarios of load migration and offloading, a higher throughput means that more load can be migrated within a unit of time.

Figure 7 demonstrates the delay performance of different service migration strategies. Due to the fact that the non-recurrent network only has neighborhood observations, it will make poor decisions during routing and is likely to select edges with large delays. This will lead to a slow

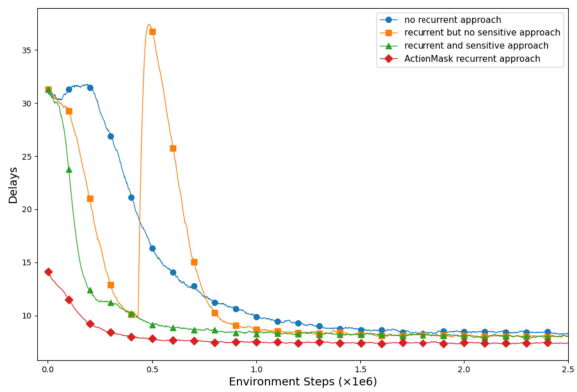


Fig. 7. Comparison of delays under different routing policies.

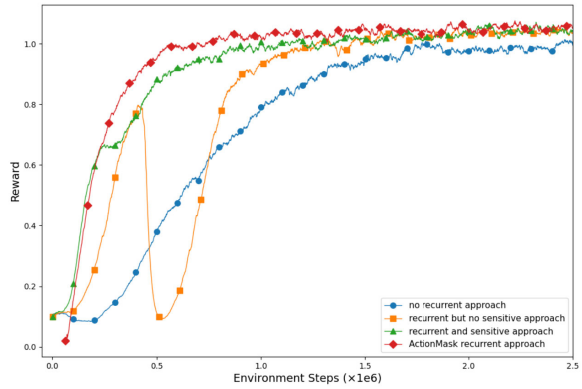


Fig. 8. Comparison of reward under different routing policies.

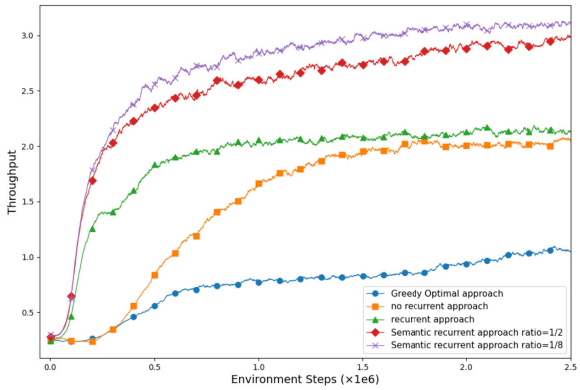


Fig. 9. Throughput comparison after semantic feature extraction.

migration process during load migration and affect the overall performance of the system.

Figure 8 shows the reward convergence performance of different service migration strategies. The figure shows that the recurrent network strategy approach is able to converge quickly. In the scenarios of load migration and offloading, a service migration strategy with rapid convergence can find the optimal migration path more quickly.

Since semantic feature extraction can compress the amount of transmitted data, the semantic information about the topological map of the satellite-UAV network is considered to be transmitted in the message recurrent delivery network. The simulation results are shown in Figure 9, Figure 10, and Figure 11.

Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.

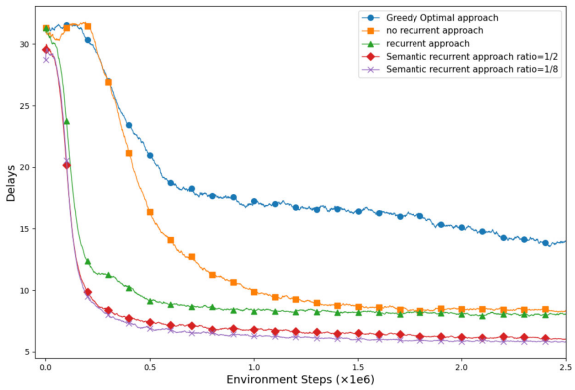


Fig. 10. Delays comparison after semantic feature extraction.

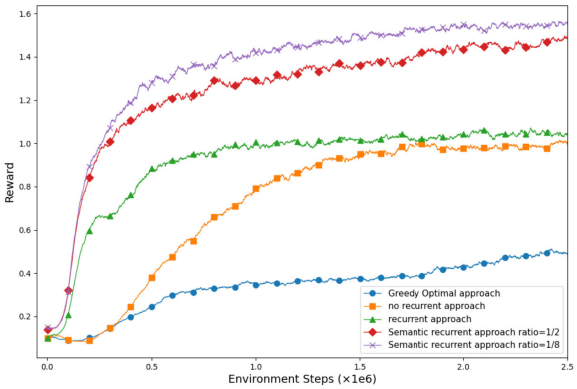


Fig. 11. Reward comparison after semantic feature extraction.

Figure 9 demonstrates the trend of throughput after adding semantic extraction. Due to the reduction in the amount of data passed in messages, the packets transmitted between satellites on the edge are less likely to block, which in turn leads to a growth in the quantity of packets arriving at the destination node. For load migration, this enables more efficient transmission of load data to the target node, thus improving the overall data migration efficiency.

Figure 10 shows the trend of delay after adding semantic extraction. After semantic extraction, the information that can be utilized between the satellite and UAV nodes to make correct service migration decisions is lighter and can be used to make decisions in fewer time steps.

Figure 11 demonstrates that the reward value can converge quickly after adding semantic extraction and the converged reward value is much higher than the corresponding reward value of other routing strategies. This indicates that semantic extraction helps the agents learn optimized routing strategies more efficiently, enabling them to achieve better decision-making benefits in scenarios of service migration with request routing.

VI. Conclusion

In this paper, we presents a novel service migration with routing framework leveraging MARL within satellite-aided UAV networks. The service migration with routing is modeled as graph representations and MARL control process, aimed at minimizing transmission delays during load migration. To address the dynamic nature of satellite-UAV network loads, we make use of a discrete denoising diffusion model to produce network topology graphs and propose a service migration strategy incorporating semantic graph extraction. The simulation results indicate that our method significantly reduces average transmission delays, enhances throughput during network service migration in satellite-UAV networks. While the current research emphasizes collaborative service migration with routing between satellites and UAVs, future investigations could extend this framework to include more heterogeneous nodes, such as ground base stations and high-orbit satellite, thereby establishing a multi-level cross-layer collaborative framework.

References

[1] Y. Jiang et al., "Integrated sensing and communication for low altitude economy: Opportunities and challenges," *IEEE Commun. Mag.*, vol. 63, no. 12, pp. 72–78, Dec. 2025.

[2] X. Qin, T. Ma, Z. Tang, X. Zhang, H. Zhou, and L. Zhao, "Service-aware resource orchestration in ultra-dense LEO satellite-terrestrial integrated 6G: A service function chain approach," *IEEE Trans. Wireless Commun.*, vol. 22, no. 9, pp. 6003–6017, Sep. 2023.

[3] W. Yuan et al., "From ground to sky: Architectures, applications, and challenges shaping low-altitude wireless networks," 2025, *arXiv:2506.12308*.

[4] J. Zheng et al., "Trust management of tiny federated learning in Internet of unmanned aerial vehicles," *IEEE Internet Things J.*, vol. 11, no. 12, pp. 21046–21060, Jun. 2024.

[5] K. Peng et al., "Delay-aware joint microservice deployment and request routing in multi-edge environments based on reinforcement learning," *IEEE Trans. Netw. Service Manage.*, vol. 22, no. 4, pp. 2971–2986, Aug. 2025.

[6] C. Xie et al., "Disrupting vision-language model-driven navigation services via adversarial object fusion," 2025, *arXiv:2505.23266*.

[7] Y. Chen, J. Zhao, Y. Wu, J. Huang, and X. S. Shen, "Multi-user task offloading in UAV-assisted LEO satellite edge computing: A game-theoretic approach," *IEEE Trans. Mobile Comput.*, vol. 24, no. 1, pp. 363–378, Jan. 2025.

[8] M. Giordani and M. Zorzi, "Non-terrestrial networks in the 6G era: Challenges and opportunities," *IEEE Netw.*, vol. 35, no. 2, pp. 244–251, Mar. 2021.

[9] A. Li, T. Zhou, T. Xu, Y. Ouyang, H. Hu, and C. Wu, "LEO satellite assisted edge computing with latency and energy optimization," *IEEE Trans. Netw. Sci. Eng.*, vol. 12, no. 4, pp. 2640–2653, Jul. 2025.

[10] C. Li, Y. Gan, Y. Zhang, and Y. Luo, "A cooperative computation offloading strategy with on-demand deployment of multi-UAVs in UAV-aided mobile edge computing," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 2, pp. 2095–2110, Apr. 2024.

[11] S. Zhang et al., "GRLR: Routing with graph neural network and reinforcement learning for mega LEO satellite constellations," *IEEE Trans. Veh. Technol.*, vol. 74, no. 2, pp. 3225–3237, Feb. 2025.

[12] J. Zheng, B. Du, H. Du, J. Kang, D. Niyato, and H. Zhang, "Energy-efficient resource allocation in generative AI-aided secure semantic mobile networks," *IEEE Trans. Mobile Comput.*, vol. 23, no. 12, pp. 11422–11435, Dec. 2024.

[13] R. Gao, Y. Xu, H. Li, Q. Zhang, and Z. Yang, "Semantic-aware jointed coding and routing design in large-scale satellite networks: A deep learning approach," *IEEE/ACM Trans. Netw.*, vol. 32, no. 6, pp. 5415–5429, Dec. 2024.

[14] J. Wang et al., "Generative AI based secure wireless sensing for ISAC networks," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 5195–5210, 2025.

[15] J. Zheng et al., "Over-the-air wireless federated learning model for generative AI," *IEEE Netw.*, early access, Mar. 12, 2025, doi: 10.1109/MNET.2025.3550959.

[16] J. Wang et al., "Generative AI for integrated sensing and communication: Insights from the physical layer perspective," *IEEE Wireless Commun.*, vol. 31, no. 5, pp. 246–255, Oct. 2024.

[17] R. Zhang et al., "Generative AI agents with large language model for satellite networks via a mixture of experts transmission," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 12, pp. 3581–3596, Dec. 2024.

Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.

[18] J. Wang et al., "Generative AI enabled robust data augmentation for wireless sensing in ISAC networks," *IEEE J. Sel. Areas Commun.*, early access, Sep. 24, 2025, doi: 10.1109/JSAC.2025.3613672.

[19] J.-H. Chen, W.-C. Kuo, and W. Liao, "SpaceEdge: Optimizing service latency and sustainability for space-centric task offloading in LEO satellite networks," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 15435–15446, Oct. 2024.

[20] Z. Li, H. Zhang, C. Liu, X. Li, H. Ji, and V. C. M. Leung, "Online service deployment on mega-LEO satellite constellations for end-to-end delay optimization," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 1, pp. 1214–1226, Jan. 2024.

[21] Y. Liu, P. Lin, M. Zhang, Z. Zhang, and F. R. Yu, "Mobile-aware service offloading for UAV-assisted IoV: A multiagent tiny distributed learning approach," *IEEE Internet Things J.*, vol. 11, no. 12, pp. 21191–21201, Jun. 2024.

[22] N. Lin, X. Han, A. Hawbani, Y. Sun, Y. Guan, and L. Zhao, "Deep reinforcement learning-based dual-timescale service caching and computation offloading for multi-UAV assisted MEC systems," *IEEE Trans. Netw. Service Manage.*, vol. 22, no. 1, pp. 605–617, Feb. 2025.

[23] N. Zhao, Z. Ye, Y. Pei, Y.-C. Liang, and D. Niyato, "Multi-agent deep reinforcement learning for task offloading in UAV-assisted mobile edge computing," *IEEE Trans. Wireless Commun.*, vol. 21, no. 9, pp. 6949–6960, Sep. 2022.

[24] F. Li et al., "Multi-UAV hierarchical intelligent traffic offloading network optimization based on deep federated learning," *IEEE Internet Things J.*, vol. 11, no. 12, pp. 21312–21324, Jun. 2024.

[25] C. Zhang and J. Yang, "An energy-efficient collaborative offloading scheme with heterogeneous tasks for satellite edge computing," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 6, pp. 6396–6407, Nov. 2024.

[26] K. Peng, L. Wang, J. He, C. Cai, and M. Hu, "Joint optimization of service deployment and request routing for microservices in mobile edge computing," *IEEE Trans. Services Comput.*, vol. 17, no. 3, pp. 1016–1028, May 2024.

[27] K. Poularakis, J. Llorca, A. M. Tulino, I. Taylor, and L. Tassiulas, "Service placement and request routing in MEC networks with storage, computation, and communication constraints," *IEEE/ACM Trans. Netw.*, vol. 28, no. 3, pp. 1047–1060, Jun. 2020.

[28] R. Li, Z. Zhou, X. Zhang, and X. Chen, "Joint application placement and request routing optimization for dynamic edge computing service management," *IEEE Trans. Parallel Distrib. Syst.*, vol. 33, no. 12, pp. 4581–4596, Dec. 2022.

[29] O. Houidi et al., "Constrained deep reinforcement learning for smart load balancing," in *Proc. IEEE 19th Annu. Consum. Commun. Netw. Conf. (CCNC)*, Jan. 2022, pp. 207–215.

[30] J. Song et al., "Trustworthy and load-balancing routing scheme for satellite services with multi-agent DRL," in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, Jan. 2023, pp. 1–6.

[31] R. Kołakowski, S. Kukliński, and L. Tomaszewski, "Hierarchical deep reinforcement learning-based load balancing algorithm for multi-domain software-defined networks," in *Proc. IFIP Netw. Conf. (IFIP Networking)*, Jun. 2024, pp. 607–612.

[32] X. You, X. Li, Y. Xu, H. Feng, J. Zhao, and H. Yan, "Toward packet routing with fully distributed multiagent deep reinforcement learning," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 2, pp. 855–868, Feb. 2022.

[33] G. Wang, F. Yang, J. Song, and Z. Han, "Optimization for dynamic laser inter-satellite link scheduling with routing: A multi-agent deep reinforcement learning approach," *IEEE Trans. Commun.*, vol. 72, no. 5, pp. 2762–2778, May 2024.

[34] D. M. Casas-Velasco, O. M. C. Rendon, and N. L. S. da Fonseca, "Intelligent routing based on reinforcement learning for software-defined networking," *IEEE Trans. Netw. Service Manage.*, vol. 18, no. 1, pp. 870–881, Mar. 2021.

[35] J. Gómez-delaHiz and J. Galán–Jiménez, "Improving the traffic engineering of SDN networks by using local multi-agent deep reinforcement learning," in *Proc. Netw. Oper. Manage. Symp.*, May 2024, pp. 1–5.

[36] G. Sun et al., "Aerial reliable collaborative communications for terrestrial mobile users via evolutionary multi-objective deep reinforcement learning," *IEEE Trans. Mobile Comput.*, vol. 24, no. 7, pp. 5731–5748, Jul. 2025.

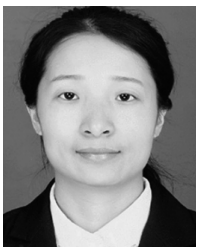
[37] X. Guo and L. Zhao, "A systematic survey on deep generative models for graph generation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5370–5390, May 2023.

[38] C. Vignac, I. Krawczuk, A. Siraudin, B. Wang, V. Cevher, and P. Frossard, "DiGress: Discrete denoising diffusion for graph generation," in *Proc. 11th Int. Conf. Learn. Represent.*, 2023, pp. 1–28.

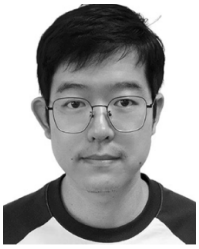
Qiongqiong Jia is currently pursuing the M.S. degree with the College of Computer Science, Northwest University, Xi'an, China. Her research interests include multi-agent reinforcement learning, service migration, and generative intelligence.

Jie Zheng received the B.S. degree in communications engineering from Nanchang University, China, in 2008, and the Ph.D. degree from the Department of Telecommunications Engineering, Xidian University, in 2014. He is currently a Professor with the College of Computer Science, Northwest University, Xi'an, China. His research interests include intelligent computing and communication, and edge intelligence.

Ling Gao received the B.S. degree in computer science from Hunan University, China, in 1985, the M.S. degree in computer science from Northwest University, Kirkland, WA, USA, in 1988, and the Ph.D. degree in computer science from Xi'an Jiaotong University, Xi'an, China, in 2005. His research interests include network security and management, and embedded internet service.



Jinping Niu (Member, IEEE) received the Ph.D. degree from Xidian University, Xi'an, China, in 2014. From 2011 to 2013, she was a Visiting Student with the School of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA, USA. She is currently a Professor with the College of Computer Science, Northwest University, Xi'an. Her research interests include wireless communications and the IoT communication technologies.



Rui Cao is currently a Lecturer with the College of Computer Science, Northwest University, Xi'an, China. He has participated in several national research projects. His relevant results have been published in findings of ACL, ACM MM, *Knowledge-Based Systems*, ICANN, MIDL, and *Heritage Science*. His research interests include neural language processing, multimodal data mining, and applications.

Jie Ren (Member, IEEE) is currently an Assistant Professor with Shaanxi Normal University, Xi'an, China. He serves as a PC member for IJCAI 2021 and SocialCom 2020 and the Session Chair for UIC 2018.

Authorized licensed use limited to: Tsinghua University. Downloaded on April 07,2026 at 08:33:16 UTC from IEEE Xplore. Restrictions apply.